

Opening the Black Box of the Deliberative Poll: The Roles of Identity and Affect¹

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Political scientists increasingly conduct Deliberative Polls. Invented by James Fishkin, the first Deliberative Poll was held in Manchester, England, in 1994 (Fishkin 1996). Since then, Deliberative Polls have been conducted many times, all over the world, and by numerous scholars.² Some concerned about citizen ignorance and apathy advocate for Deliberative Polls as a method that can lead citizens to become more engaged and knowledgeable about public affairs, noting that Deliberative Poll participation causes both changes in their opinions and learning about the topic (Luskin, Fishkin and Jowell 2002, Andersen and Hansen 2007).

Despite the increasing popularity of the method, however, scholars know very little about how Deliberative Polls actually produce these results. Certainly the collective psychology of group involvement must play a role, but the unanswered question is: How? We apply social identity theory and theories of affect to this question and argue that people become engaged because of the psychology of group dynamics. The minimal intergroup paradigm predicts that people form group identities spontaneously. Following from this, we argue that people form

² Aside from the United States, Deliberative Polls affiliated with James Fishkin have been conducted in the United Kingdom, Brazil, Bulgaria, China, Greece, Italy, Hungary, Northern Ireland, and one included participants from all European Union nations (The Center for Deliberative Democracy 2009). Political scientists not directly affiliated with Fishkin have conducted their own Deliberative Polls in Denmark (Hansen and Andersen 2004, Andersen and Hansen 2007) and Australia (Eggins et al. 2007). The Jim Lehrer Newshour held Deliberative Polls in cities across the country, conducting them in January and October of 2004, and also from 2005 to 2007 (The On-Line NewsHour 2009). In addition, for three years in a row the American Association of State Colleges and Universities has led consortiums of schools to hold their own Deliberative Polls (American Association of State Colleges and Universities 2009).

group identities with the small group members that they discuss in. We also argue that people have feelings during their discussion, and that these feelings operate both independently of, and in conjunction with, their group identities to bring engagement and attitude change.

THEORY

A Deliberative Poll begins by drawing a random sample of a population that a researcher wishes to study. Participants are asked about a public policy. At the end of the survey, participants are asked to attend a session of discussion about the topic. Typically, an incentive is offered to increase attendance. At the Deliberative Poll site, participants are randomly assigned to discussion groups of about 10 people. They read a briefing document about the topic. They are told that they will attend a panel where they can ask policy experts questions, and the focus of the small group discussions is to generate questions for the policy experts. After several sessions of discussion, all discussants meet to ask the experts questions, and this is followed by a final meeting of the small groups.

Three studies have addressed the question of how Deliberative Polls lead to attitude change. The first analyzes data from Deliberative Polls in England and finds that attitude change stems from two sources: primarily from learning about the issues, and secondarily from a group consensus effect where people's attitudes move toward that of the group average (Luskin, Fishkin and Jowell 2002). On this second point, other scholars fail to find group consensus effects (Gastil, Black and Moscovitz 2008, Farrar et al. 2009, and Petrow 2009). The second study finds that the discussions themselves lead to attitude change (Andersen and Hansen 2007).

The third study attempted to consider the relationships between attitude change, political engagement, and group identity (Eggins et al. 2007). Unfortunately, this study is fraught with errors that make any conclusions tenuous.³

However, finding that discussions, or information, lead to attitude change still presents scholars with a kind of black box of understanding regarding how these changes occur. Farrar et al. (2003) summarize this situation by writing,

“How far do the changes [in attitudes] stem from the sheer quantity of information acquired, from the degree to which it is balanced, from the social properties or intellectual content of small group discussions, from the attitudinal or socio-demographic heterogeneity of one’s fellow discussants? How far do they depend on reading versus listening? On solitary versus social processes more generally? On the adoption of “empirical premises,” which are debatable, versus the absorption of “facts,” which are not (Luskin, Fishkin, and Jowell 2002)? On learning versus thinking? On empathy or identification versus argument? It is already clear, from both earlier statistical analyses (Luskin, Fishkin, and Jowell 2002) and the experimental results above, that the quantity of factual information absorbed plays a major intervening role, but much more remains to be explored.”

In other words, the black box in which changes occur during the Deliberative Poll remains a large one. We seek here to make it a bit smaller.

The Political Psychology of the Deliberative Poll

The minimal intergroup paradigm is among the most replicated effects in the social sciences (e.g. Brewer 1979, Brown 1995). A minimal intergroup paradigm experiment involves

³ The authors measure group identity in a manner that is unrelated to either their own or any other conceptualization of identity. They measure engagement as the participants’ own subjective sense of being engaged, instead of any objective measure. They also model hypothesized causal feedback between factors inappropriately, calling into question all of the results reported by their models.

randomly assigning people to groups, and then observing people's willingness to act to benefit the in-group and derogate the out-group, as driven by group identity. The minimal intergroup paradigm predicts that people in Deliberative Polls will form bonds of attachment and affection with their small-group members. Social identity theory suggests that group identity leads people to see others as being like themselves, in the sense that people perceive the sharing of common attributes and traits with others. This will lead to what Brewer and Gardner (1996) term a sense of "we-ness"; one's sense of self is extended to include others, such that one has a sense of shared fate, or accomplishment.

Group identity, and its strength, originate from four sources: the valence of group membership, assimilation to a group prototype, value consistency, and outgroup competition or threat (Huddy 2001). We expect that most Deliberative Poll participants will have positive feelings for their group because they have personally chosen to participate in the event. We also believe that participants will enter the groups with notions of what a prototypical member would look or act likely; namely, someone who participates and cooperates in civic discussion. We believe that as small discussion groups continue with their discussions, some group members (those who participate more in a positive way) will become exemplars of the group, for other group-identifying members to assimilate to. We also see value consistency as having a lot to do with forming and strengthening group identity, as those who feel the strongest sense of civic duty, and interest in politics, will see the group's purpose as being consistent with their values. Finally, there is a lack of outgroup competition or threat between the Deliberative Poll discussion groups, which will serve to weaken the group identity.

Identities produce behaviors consistent with the shared understandings of the group's meaning to the identifiers (Huddy 2001). The purpose of the small-group discussions is to work

cooperatively to discuss the issues. Thus, the meaning of the identity to the participants should be to discuss cooperately and productively. Group identity leads to behaviors for group welfare (Brewer and Kramer 1986, Kramer and Brewer 1984) and group cooperation (Messick and Brewer 1983). Therefore, people who endorse these identities to a greater extent should be more engaged in the discussions than those with weaker identities. Identity strength should thus predict both attitude change and greater learning as side-effects of the identity-driven engagement.

Feelings toward the other group members should matter too. Some group discussants will have positive feelings for fellow group members, others may be emotionally disengaged and not feel much at all, and still others may have negative feelings. Appraisal theories of affect predict that people assess their life experiences and situations, with these appraisals resulting in emotional reactions to events that cause different behaviors and cognitive processing (e.g. Lazarus 2001). As a positive emotion, researchers find that happiness generally leads people to take action, but also to process information heuristically. The political scientists Marcus, MacKuen and Neuman (2000) find that enthusiasm leads to greater campaign interest, and also affects other measures of political engagement, such as caring who wins an election. Enthusiasm is also associated with turning out to vote. Psychologists find that happiness leads to social engagement, such as helpfulness, generosity, responsibility and friendliness or sociability (see Isen 1987 for a review).

However, anxiety is the emotion associated with learning during a campaign, not enthusiasm. This replicates the finding in psychology that positive emotions lead people to rely on their pre-existing beliefs, processing heuristically (see Lerner, Tiedens, Gonzalez 2005 for a review). Despite this, other psychologists find a role for positive emotion in processing.

Frederickson (2001, 2004) reviews the literature and finds that positive emotion's role in thinking makes it more: Unusual, creative, flexible, open to information, and efficient – in essence, broadening cognition. Alice M. Isen's (2000) review of the happiness and processing literature explains what is meant by efficient thinking when she argues that happiness leads to more heuristic processing, but also more learning. These dual processes result from a cognitive integration that maximizes reliance on existing knowledge structures, but also motivates learning information about unknowns. Importantly, happiness does not increase processing universally, but only selectively. People only process information that is consistent with their positive mood, to protect it, unless they have an obligation or good reason to process negative information (e.g. Isen and Simmonds 1978). This suggests that happiness should lead to greater learning and processing about topics that maintain the happiness, unless individuals are provided with a good reason to do otherwise. In a Deliberative Poll, it's not clear what those good reasons would be.

While the findings on happiness and learning are mixed, the finding that anxiety leads to learning is more consistent. Anxiety leads to heightened focus and attention to threatening stimuli. This can lead to learning. Political scientists find this effect in political campaigns (Marcus, Neuman and MacKuen 2000; MacKuen, Marcus, Neuman and Keele 2007), when people search the Internet during political campaigns (Valentino, Hutchings, Banks and Davis 2008; Valentino, Banks, Hutchings and Davis 2009), and to a limited extent during the run-up to the War in Iraq (Huddy, Feldman and Cassese 2007). However, the role of anxiety is clearly, to some extent, contextual. People anxious about a presidential candidate are more interested in the campaign, but only those people high in internal efficacy (Rudolph, Gangl and Stevens 2000). Anxiety may only promote learning if people believe they can overcome the threats they face (Nadeau, Niemi and Amato 1995). Too much anxiety can limit processing, and learning,

because people engage in avoidance behaviors (LeDoux 1996, Panskepp 1998). After people have chosen a candidate in an election, anxiety about that candidate only leads to more learning when people discover unsettling information about the person, and then the learning only occurs about that candidate (Redlawsk, Civettini and Lau 2007). In the aftermath of 9/11, the anxious did not learn more about the situation compared to the non-anxious (Feldman and Huddy 2005).

Finally, we conclude our discussion of affect by considering another negative emotion – anger. Anger holds much in common with positive emotions such as happiness. Anger leads to engagement (Carver 2004, Berkowitz and Harmon-Jones 2004, Harmon-Jones and Allen 1998, Harmon-Jones and Sigelman 2001, Lerner and Keltner 2000, Mackie et al. 2000). Anger causes less cognitive processing and a greater reliance on pre-existing beliefs (e.g. Tiedens 2001).

While anger may lead to less learning, it does lead to attitude change. Conover and Feldman (1986) find that anger toward President Reagan, and not anxiety, led to lower Presidential approval. Also, anger led to less tolerance after 9/11 (Skitka, Bauman and Mullen 2004). While happiness leads to attitude change in support of an attitude object, anger leads to attitude change in opposition. Anger, then, leads to engagement in the form of action, but not engagement in the form of careful learning and processing. If anger does lead to processing, it is not the thoughtful and careful variety, but instead the swift and effortless variety.

This discussion leads us to hold several expectations as to how identity and affect will lead to learning and attitude change during a Deliberative Poll. First, we expect that strong group identities will lead to a deeper engagement with the group. This will lead to greater learning, as well as greater attitude change. Second, we expect that anxiety felt during

discussions will lead to greater learning, as well as greater attitude change.⁴ Third, we predict that anger will lead to attitude change, but not to greater learning.⁵ Fourth, we hypothesize that happiness felt toward other discussants will lead to attitude change, but we are agnostic as to whether or not it will lead to greater learning given the uncertain connection between happiness and learning presented in the literature.⁶

⁴ This raises a question of what would cause discussants to feel anxious during the discussions. Appraisal theories of affect suggest that anxiety arises when people feel threat but are uncertain how to respond, or uncertain if they can respond (Lazarus 1991, Weiss et al. 1999). In the context of the Deliberative Poll, perhaps people feel challenged by other participants, and don't know how to respond. Perhaps people seek standing in the group (respect) and are uncertain if they are achieving it, or did achieve it. Perhaps people try to persuade others, but are unclear if they succeed.

⁵ Anger typically results when someone who feels powerful or capable has a goal blocked (Lazarus 1991, Weiss et al. 1999, Carver 2004). In the Deliberative Poll context, this could be the case when someone is attempting to make an argument which is then challenged or refuted by others. Someone who is argumentative, or skilled at arguing, might feel angry.

⁶ The discussions of happiness and anger predict a direction of the attitude change. However, the affect here is general in nature, and so notions of direction do not apply. All three emotions -- happiness, anxiety, and anger -- predict engagement in the task at hand. Happiness should lead to a relaxed and jovial kind of openness and engagement, anxiety to a kind of on-edge agitation, and anger to a kind of seething agitation. All three types of affect predict that emotional engagement with other group members leads to attitude change because they should all heighten processing (although they do so in different ways).

How Affect Interacts with Identity, and How Affect Interacts with Affect

We also expect that the effects of the different emotions will intensify as group members hold stronger small group identities. In other words, we predict the presence of interaction effects. Social psychologists find that people with strong group identities feel stronger group-based emotions (Branscombe and Wann 1991, McCoy and Major 2003, Doojse et al. 1998). Political psychologists find that group-identifying whites are more likely to vote in Congressional elections with black candidates when they feel stronger racial affect (Petrov forthcoming). This indicates that for group identifiers, the group-related emotions have greater meaning to them, compared to those who are not group identifiers (or with weaker identities). In the case of a Deliberative Poll, this would be the case because part of what motivates the small group identity is discussants' support of the Deliberative Poll experience, perhaps as something related to civic participation. This value-consistency motivation, then, will lead to happiness if one's desires and expectations are met, but anger or anxiety if they are not. With all three emotions, the affect will have more meaning to the group identifiers because the identity and feelings are rooted in a motivation to achieve something in a Deliberative Poll (see Garcia-Prieto and Scherer 2006 for a discussion of how identity and affect are both based in people's motivations).

Thus, we expect that as group members hold stronger identities, happiness, anxiety and anger will all lead to greater attitude change. This would be the case because all three emotions predict engagement in the experience, and the emotions should serve as even greater motivators as discussants hold stronger identities. We also expect that anxiety towards group members will interact with identity to lead to greater learning. We are agnostic as to whether identity will interact with happiness toward group members to predict greater learning.

Finally, we hypothesize that feelings during the Deliberative Poll will act together with other feelings to intensify discussants' group experiences. This will lead to changes as well. Differential Emotions Theory (DET) predicts that people typically react in situations with multiple emotions, and that these emotions interact with one another to influence people's behaviors (Izard, Ackerman, Schoff, and Fine 2002). They react by either amplifying or attenuating the effects of the other emotions. DET predicts that if the properties of the emotions are the same, then they reinforce each other, but if they are in conflict, then the effects are attenuated. For example, anger and anxiety felt in isolation should both lead to attitude change, so together they should lead to even more. However, the anger should reduce the learning, or careful processing, that the anxiety leads to. We also expect anxiety and happiness will operate together. We see this combination leading to attitude change, and we are uncertain as to how it will affect learning.

METHODOLOGY

We conducted a Deliberative Poll in 2007 at a large, public university in the Midwest. We sent the initial pre-event survey to all undergraduate students at the university over email, and 40 attended the Deliberative Poll. The topic was immigration reform. The Deliberative Poll took place over one day, from 10 a.m. to 3 p.m. There was a briefing document which participants read before the discussions began, and it covered legal and "unauthorized" immigration, as well as immigration law. There was a plenary panel of experts on immigration, which featured two professors from the university, and a graduate student.

We measured small group identity by creating a scale from three items. These were the questions: How much did you feel that the people in the group were people like you, how close did you feel to the people in the group, and how much did you feel that you had a lot in common

with the people in the group? Each question had four response categories. The Cronbach's alpha is .78.

We also measured happiness, anxiety and anger felt during the discussions. We measured affect with two items: how [enthusiastic, angry or anxious/nervous] did you feel while participating in the group discussion; and, how [happy, angry, or anxious/nervous] did the people in the group make you feel. Each question had four response categories. The Cronbach's alpha for the happy/enthusiasm scale is .54, the anxiety scale is .81, and the angry scale is .91.

We report the results for four dependent variables: the amount of learning that took place during the Deliberative Poll, how many comments people typed about the immigration issue, policy polarization, and polarization on negative immigrant attitudes. We also measured the degree to which each small discussion group's preferences were single-peaked, but the results are inconclusive in this regard, and we do not report them.

The Deliberative Poll questionnaire contained the following factual questions which were covered in the briefing document. The questions were asked before the event, and again after. The questions were: About what percent of people currently living in the United States were born outside the United States; The current number of unauthorized immigrants living in the [state in which the Deliberative Poll was conducted] today is estimated to be; As far as you know—is a child of an unauthorized immigrant who is born in the United States a U.S. citizen; As far as you know, are public elementary and secondary schools in [the state] required to teach the children of unauthorized immigrants who live in [the state]; and, As far you know, are unauthorized immigrants eligible to receive federally- funded welfare benefits. The responses to these questions compose an index, ranging from zero (for none correct) to five (for all correct). The average number of correct increased from 3.15 before the Deliberative Poll to 4.15 after.

We allowed people several opportunities to type responses to open-ended questions. People were asked: What, if any, are your other major concerns about immigration of unauthorized or undocumented immigrants? We asked participants this question both before and after the Deliberative Poll occurred, and we used the independent variables to predict whether or not someone typed any comments. We include this variable as a dependent variable in our analysis as a gauge of engagement in the Deliberative Poll.

We also created a scale to measure policy polarization. We began by creating a scale out of two questions that reflected favoring: making it illegal for employers to hire unauthorized immigrants, and increasing penalties for employers who hire unauthorized immigrants. We found the Cronbach's alphas to be .89 at time one and .76 at time two. Each question had five response categories (from strongly favor to strongly oppose), and we defined the middle category (three) as those who responded that they lacked an opinion. We then took the absolute value of the difference between each response and three. Higher values, then, indicate greater distance from the middle response (not having an opinion), indicating greater movement at time two, compared to time one, away from the middle of the scale.⁷

We calculated a similar polarization scale for negative attitudes toward immigrants. We measured negative attitudes toward immigrants with eight questions measuring agreement that: new immigrants joining the workforce drive down wages, that immigrants add to the crime problem; and, the beliefs that legal or illegal immigration (with different questions for both topics) -- hurt American jobs, increase the danger of terrorism, and contribute to crime. The

⁷ The group-level average for polarization decreased from 1.11 at time one to 1.04 at time two (a statistically insignificant change). However, the correlation between the times one and two responses is .32, indicating a lot of individual-level change.

Cronbach's alpha at time one was .91. and .86 at time two. The response categories were similar to the policy scale described above, and we followed the same procedure to build the folded negative immigrant attitude polarization scale.⁸

We analyzed these scales as dependent variables, estimating a logit model to predict whether or not discussants wrote comments (because of the dichotomous dependent variable), and OLS regressions for all others. We want to capture which factors lead to change from the beginning of the Deliberative Poll to the end, and so we include lagged dependent variables in all of the models.

RESULTS

We begin reporting results in Table 1. Table 1 reports the results we found for the effects of the identity and affect scales on participants' behaviors. Two of the analyses are OLS regression results for changes in knowledge, and the third result is from a logit analysis predicting whether or not people typed comments at time two concerning their thoughts regarding unauthorized immigration, controlling for whether or not they typed comments at time one.

[Table 1 about here]

We begin with the first column, column A, which is an additive model to predict increases in knowledge at time two. Our inclusion of the lagged dependent variable means we should interpret the independent variables as change scores, or predicting change in knowledge from before the Deliberative Poll, to after (e.g. Kmenta 1997). We find that the happiness and

⁸ The group-level average for polarization increased from .88 at time one to .90 at time two (a statistically insignificant change). The correlation between the times one and two responses is .56, indicating a fair amount of individual-level change over time.

anxiety scales both predict greater learning ($b=.54$, $p<.01$; and, $b=.47$, $p<.01$, respectively).

Anger is negative, but very close to zero, while identity is also negative. However, neither result is statistically significant. We also find that more years in attendance at college predicts greater learning during the event ($b=.39$, $p<.01$).

We plot the changes in the predicted knowledge score in Figure 1. The y-axis reflects the predicted value for the number of correct answers participants gave at time two (after the Deliberative Poll). The x-axis has two categories: the predicted value of knowledge for the lowest scores on the happiness and anxiety scales, and the predicted score for the highest scores. The values of the other variables in the model (including the lagged dependent variable) are all set at their means. One can see that the increase in the predicted score for the number of correct answers associated with happiness, from about three to almost five, is statistically significant at the $p<.05$ level (this is not the case for the increase for the anxiety scale). This statistically significant increase indicates that when group members reported feeling the most happiness they, on average, answered two more questions correctly after the Deliberative Poll than before.⁹

⁹ The predicted score here is the predicted score for knowledge at time two. However, one of the factors predicting this score is knowledge at time one. As such, the coefficients used to estimate the predicted value are predictors of change over time, from time one to two, for the dependent variable. Thus, there are two ways in the model in which time operates: First, the lagged dependent variable reflects how values of the dependent variable at time one affect values at time two; and second, all of the other independent variables predict change in the dependent variable from time one to time two. As such, the predicted score for knowledge, given happiness and anxiety set at their lowest values, is determined by: Someone's average knowledge at time one; the average values on all of the other independent variables (which predict how knowledge from

[Figure 1 about here]

While the results from the table for happiness and anxiety are statistically significant, the only emotion to increase the predicted number of correct responses that is statistically significant is for the degree of happiness or enthusiasm people felt as they participated. This indicates that, at least in this Deliberative Poll, enthusiasm and happiness played a greater role in learning than did anxiety, contradicting a finding that is dominant in political science (e.g. Marcus, Neuman and MacKuen 2000), but that is contested in social psychology (e.g. Isen 2000). It appears that in this context happiness did motivate participants to learn more about what they didn't know, while the evidence is less clear in this regard for anxiety.

We return to Table 1 to consider a result for an interaction between two emotions. We estimated models with interaction effects between anxiety and anger, and anxiety and happiness, and we report the result for happiness interacted with anxiety in column B. The interaction effect itself is negative, but not statistically significant. However, the component effects are. These represent the effects of the emotions when the values of the other emotions are at their lowest levels. Anxiety increases learning when happiness is at its lowest level ($b=1.89, p<.01$), and happiness increases learning when anxiety is at its lowest level ($b=1.27, p<.01$). An

time one to time two changes); and, happiness and anxiety set at their lowest levels. However, for the highest scores of happiness and anxiety, the increases in knowledge seen in Figure 1 reflect the changes in the predicted score at time two, relative to time one. This is the case because the coefficients that generate the increases in the predicted score reflect how these factors are related to the change in knowledge, from before the event to after. This means that the increase in the predicted score is the change from before the event to after that is associated with these variables.

additional year of schooling also increases learning ($b=.36, p<.05$). We plot the results for affect in Figure 2 to ascertain how these factors change the predicted value of knowledge.

[Figure 2 about here]

The figure reveals that the interaction between anxiety and happiness results in a very modest increase in knowledge. The change is not statistically significant. However, the increases for the two factors are both large and statistically significant ($p<.05$). Moving from the lowest score on happiness, to the highest score, increases the predicted value of correct answers from about three to seven. This result is out of the bounds of the scale, which has a maximum score of five. However, the result suggests that feeling the most happiness, without any anxiety, caused people to answer four additional questions correctly after the Deliberative Poll, compared to before it. The increase for anxiety is from about three to almost eight, which indicates moving from answering no questions correctly before the event, to answering all of them correctly after. Happiness and anxiety together slightly increase knowledge, but in isolation, both are strongly related to learning during the Deliberative Poll. This replicates the earlier result from the additive model (column A), but in this case the effects are larger because happiness and anxiety are not being “held constant” statistically; instead, their influence is removed from the other emotion as a side-effect of the interaction term. Happiness and anxiety, then, experienced in their pure forms, produce the largest effects. This contradicts a prediction of DET, which is that the emotions should act together to amplify the effects on learning. While happiness and anxiety both lead to greater learning, they do so most powerfully in the absence of the other emotion, not together with the other emotion.

The final result from Table 1 concerns factors predicting whether or not discussants typed comments about unauthorized immigration.¹⁰ In column C, we report results from a logit regression. The model is additive, with happiness resulting in fewer comments ($p < .01$), and anxiety and group identity more comments (both $p < .01$). The coefficient for anger is negative but not statistically significant. Three of the control variables are statistically significant: sex (males are coded as high, $b = -6.89$, $p < .01$), age ($b = .58$, $p < .05$), and party identification (Strong Democrats are coded as high, $b = -1.02$, $p < .05$). Because the coefficients are in the units of the log of odds ratios, they are not intuitively interpretable, and so we plot the changes in the predicted probability of leaving a comment for the factors. We report these changes in Figure 3.

[Figure 3 about here]

While we reported earlier that happiness results in more engagement in the form of greater learning, here it is associated with withdrawing, in the form of a lower predicted probability of leaving a comment. The change in the predicted probability, however, is not statistically significant. The increases for anxiety and group identity are both statistically significant ($p < .05$), with both factors increasing the predicted probability of leaving a comment from about 20% to about 100%. This indicates an increase of about 80% in the predicted probability of leaving a comment, from before the Deliberative Poll to after it. While group identity did not predict engagement in the form of learning, here it predicts engagement in the form of leaving a comment. Anxiety's positive association with learning in the column A model did not lead to a significant increase in the predicted score for knowledge, but anxiety does lead to a statistically significant increase in this form of engagement.

¹⁰ We also estimated models predicting the number of comments people typed, but our variables of interest were unrelated to that dependent variable.

We now turn to Table 2, in which we report results for changes in attitudes from before the Deliberative Poll, to after. All of the additive models with these dependent variables yielded statistically insignificant results, and so we only report results with interaction terms. In column D we present the results for the interaction between group identity and anxiety on negative immigrant attitude polarization.¹¹ The interaction effect is negative and statistically significant, predicting that a one-unit increase in both variables results in movement toward the middle of the scale ($b = -.39$, $p < .01$). The main effects of the interaction are both positive and significant: Group identity without anxiety leads to greater polarization ($b = .74$, $p < .01$), as does anxiety ($b = 1.15$, $p < .01$). We also find that anger leads to less polarization ($b = -.11$, $p < .10$).

[Table 2 about here]

We turn to Figure 4 to see the effects of the interaction and its components on the predicted score of the negative immigrant attitude polarization scale. Despite the fact that the interaction coefficient is negative, the interaction actually increases the predicted score on the scale, from about .5 to a bit more than one. This very modest increase is not statistically significant. However, the two component factors are: anxiety and group identity both increase negative immigrant attitude polarization ($p < .05$). Group identity begins at about one and increases to about three (for a two-category increase from before to after the poll), while anxiety begins at about one and increases to 3.75 (for an almost three category increase). Theory clearly predicts that anxiety should have greater effects as group identity strengthens. While the modest

¹¹ We also measured total attitude change, which was the absolute value of the difference between each individual's dependent variable score at time one and time two. We did so for both of the attitude scales that we analyze here. We found the independent variables we consider here to be unrelated to those dependent variables.

and statistically insignificant increase in the predicted score for the interaction does provide a bit of evidence this is the case, the impacts of anxiety and group identity are clearly much larger when they operate singularly. The components of the interaction tell us that the anxiety discussants felt among those lacking group identity significantly increased attitude polarization, as did group identity among those reporting no anxiety. In other words, feeling anxious during the discussions mutes the impact of group identity, and feeling identity with the group during discussions mutes the impact of anxiety.

[Figure 4 about here]

We return now to Table 2, reporting the results for the next model in column E. Negative immigrant attitude polarization remains our dependent variable, and we find that the interaction between happiness and anxiety moved discussants toward the middle of the scale (i.e. in a less polarized direction; $b = -.28$, $p < .05$). The effect of happiness is positive but not statistically significant, while the effect of anxiety is positive and statistically significant ($b = .87$, $p < .05$). We parse these effects out more thoroughly in Figure 5, where we plot the changes to the predicted score.

[Figure 5 about here]

The interaction effect between happiness and anxiety decreases the predicted score of the negative immigrant attitude polarization scale from .5 to just below zero (a result which is both out the bounds of the scale and not statistically significant). Happiness, among those reporting no anxiety, increases the predicted score from about .75 to about 1.6 (this result is not statistically significant). Finally, anxiety, among those reporting no happiness, raises the predicted score from .75 to 3, a statistically significant increase ($p < .05$). We predicted that the interaction between happiness and anxiety would increase attitude polarization. We found little

relationship between this interaction and attitude polarization; however, anxiety, purified of any feelings of happiness, does increase attitude polarization.

Our final dependent variable is for the policy difference polarization scale, the results for which we report in Table 2. In column F, we find that the interaction between anger and group identity predicts greater policy difference polarization ($b=.40, p<.05$). Age also increases policy difference polarization ($b=.05, p<.01$). To gauge how the anger and group identity interaction affects the predicted score for the scale, we plot the results in Figure 6.

[Figure 6 about here]

We find in Figure 6 that the interaction increases the predicted score for the policy difference polarization from .75 to about 2.25. The result is statistically significant ($p<.05$). Anger and group identity alone both decrease the predicted score, with anger doing so more than identity, but neither result is statistically significant. In this case, our result is consistent with our theoretical predictions: group identity amplifies the meaning of the anger felt during the discussions, increasing policy attitude polarization.

Our final result is for the interaction between anger and anxiety, which we report again in Table 2 (this time in column G). The term for the interaction between anger and anxiety indicates an increase in policy polarization ($b=.46, p<.10$). The result for the control variable age is also statistically significant ($b=.04, p<.05$). We turn to Figure 7 to ascertain if the interaction results in an increase in the predicted score for policy polarization.

[Figure 7 about here]

The anger and anxiety interaction increase the predicted score of policy polarization from about .5 to almost 2 ($p<.05$). The components of the interaction term, anger and anxiety, are associated with statistically insignificant decreases in the predicted score of policy polarization.

As with the previous result, this one, too, is consistent with our theoretical expectations. We predicted that anger and anxiety would act together to increase attitude polarization, and we find this to be the case.¹²

We have evaluated the coefficients from these models and found evidence that identity and affect associated with the small discussion groups in a Deliberative Poll lead to both engagement and attitude change. We turn to another analytic technique to evaluate this question as well – comparing a measure of model fit with the identity and affect variables included, to a measure of model fit produced after they are removed. We report these results in Table 3.

[Table 3 about here]

Our first dependent variable is the change in knowledge from before the Deliberative Poll to after. We find that a model without group identity, happiness, anxiety and anger (but with the other variables included in Table 1) produces an R-squared statistic of .16. When we estimate the model we report in column A, the R-squared more than doubles to .36. Conducting an F-test for statistical significance, we find the increase to be significant at the $p < .10$ level.

The next dependent variable of interest is for whether or not the discussants typed comments. The model is a logit, and so the measure of model fit we use is the Wald Chi-squared statistic. The basic model produces a statistic of 16.58. The model we report in column B produces a statistic of 18.95. The difference is not statistically significant.

The third dependent variable of interest is the negative immigrant attitude polarization scale. The basic model already explains a good deal of the variance of the dependent variable

¹² To ascertain if learning had effects on the other dependent variables among those with stronger identities and affects, we estimated interaction effects between learning and the other factors of interest. The results were inconclusive (not reported but available on request).

(46%). Adding the affect and identity variables in an additive fashion increases the R-squared to .51 (we do not report the coefficients for this model, and the increase is not statistically significant). However, the model we report in column D produces an R-squared of .58, which is a statistically significant increase over the basic model at the $p < .10$ level.

The final dependent variable of interest is the policy difference polarization scale. The basic model produces an R-squared of .27. When we add the identity and affect variables in an additive fashion, the model fit improves to .37 (we do not report the coefficients from this model, and the increase is not statistically significant). However, the model we report in column F has an R-squared of .45. This doubles the size of the R-squared from the basic model, and the increase is significant at the $p < .10$ level. In conclusion, for learning and attitude polarization during the Deliberative Poll, including the identity and affect variables leads the models to enjoy a better statistical fit with the data.

DISCUSSION

What do these results tell us about the method of Deliberative Polling? Much of the work of the Deliberative Poll takes place in the small discussion groups. The method is designed so that participants spend almost all of their time in those groups. Because the small groups are so central to the method, we find that many of the effects observed in a Deliberative Poll are due to the small group identities and feelings of the discussants. In other words, identity and affect matter for behaviors and attitudes. However, the relationships are more complicated than being merely additive. They are more conditional. Typically, identity interacts with affect, and affect interacts with affect, to produce the changes observed in Deliberative Polls.

We note that these findings stem from a Deliberative Poll that was different from a typical Deliberative Poll (as conducted by Fishkin and his associates) in at least three ways.

First, it was shorter. Typical Deliberative Polls last one full day, and this one lasted only five hours. Second, a typical Deliberative Poll will involve many more participants. It is common to have hundreds, and this one only had forty. Third, a typical Deliberative Poll will be covered by television reporters. This Deliberative Poll was not covered by the media as it unfolded.

These differences suggest that, because of the shortness of time, as well as the lack of media coverage, the effects of these factors in a typical Deliberative Poll might be greater. Certainly, as people discuss for longer periods of time in their small groups, their identities and feelings may become more intense, and/or more salient, thus increasing their effects. The media coverage may have the same effect, leading participants to feel that more is at stake because the event is “important,” which would bring a seriousness and intensity that could heighten the effects of identity and emotion. Finally, the small number of participants makes it harder for the models to find statistically-significant effects.

If one wanted to gain a better understanding of how identity and affect shape these discussions, one could experimentally manipulate these variables. There are a myriad of possible ways to do so, and we only suggest a few here. One simple way would be to randomly assign some of the discussion groups names, or symbols that stand for the groups. Alternatively, some kind of competitive process could be established between groups to heighten out-group derogation, which should dramatically strengthen the identities.

Finally, we note that these findings tell us not only about small discussion groups in Deliberative Polls, but also about small discussion groups in general. Any time a small group of people meet to discuss public affairs, one can imagine that identities and feelings will form, influencing people’s attitudes and participation as discussion progresses. Of course, the dynamics of identity and affect will be different outside of a Deliberative Poll than inside of one.

We see our analysis here as suggesting several research questions for future exploration. First, how does the length of a Deliberative Poll affect the group identities and feelings? Second, what are the determinants of the group identities and feelings? For example, one determinant of the group identity should be value-consistency, such that people who are more civically-oriented should hold stronger identities. Third, experimentally-manipulating identity will lead to cleaner estimates of the effects of identity on the behaviors and attitudes we study here.

CONCLUSION

Based on our results from this single Deliberative Poll, the relationship between people's emotional experiences during the small group discussions, and their learning during the event, is strong. Happiness increases the predicted score of knowledge at the end of the Deliberative Poll. In the absence of anxiety, happiness dramatically increases learning at the event, as does anxiety in the absence of happiness. While DET predicts the strongest results should be for when they operate together, our results do not support this view. However, including the affect and identity variables did more than double the R-squared for the model, which is a statistically-significant increase over a model lacking any of the identity or affect factors ($p < .10$).

Our second measure of engagement predicted whether or not people typed comments about unauthorized immigration. Happiness did not increase the predicted probability of typing a comment, but anxiety and group identity did ($p < .05$). The measure of model fit including the identity and affect factors did increase compared to the basic model, but the increase was not statistically significant. However, anxious and group-identifying discussants were more likely to type comments at the end of the experience, compared to before it.

We also had two measures of attitude change. The first was the negative immigrant attitude polarization scale. We expected that the interactions of happiness and identity, and

anxiety and identity, would increase negative immigrant attitude polarization. This was not the case. However, as we report in Figure 4, anxiety increased polarization among those lowest in group identity, while group identity increased polarization among those lowest in anxiety. This indicates that when the factors operate together, they actually mute each others total effects, but among those reporting a total lack of the other factor, identity and anxiety do lead to either more positive or more negative views of immigrants. Also, the model with the anxiety and group identity interaction term was a statistically significant improvement in model fit over the basic model that lacked any of the identity and affect variables ($p < .10$).¹³

The second measure of attitude change was the policy difference polarization scale. We find that the feeling of anger among group identifiers increases the predicted score on the scale ($p < .05$, Figure 6). We also find that, consistent with the DET theory, the interaction of anger and anxiety increases the predicted score on the polarization scale ($p < .05$, Figure 7). For model fit, we find that the model we reported in column F of Table 2 had a statistically significant improvement of model fit over a model without any identity or affect variables ($p < .10$).

In summary, then, the small discussion group identities and affects relate to the increased learning, typing comments, and attitude polarization, which occurred during this Deliberative Poll. In two instances these effects were simple and additive. The happier people reported feeling during their discussions, the more they learned. Also, the stronger their identities, and the more anxious they felt, the greater was the likelihood they would type a comment about unauthorized immigration at the end of the event, compared to the beginning.

¹³ We also expected that the happiness and anxiety interaction would lead to greater attitude polarization. Again, as we find in Figure 5, this was not the case. However, anxiety, among those lowest in happiness, did lead to either more or less negative immigrant attitudes.

In all other instances we find the relationships between independent and dependent variables to be conditional. In Figure 3 we find that happiness and anxiety both increase the predicted score for increases in knowledge from before the event to after, but only when people felt happiness without any anxiety, and anxiety without any happiness. In Figure 4 we find that anxiety increases negative immigrant attitude polarization, but only among those lacking any group identity. We also find that group identity has the same effect, but only among those lacking any anxiety. Similarly, in Figure 5, we find that anxiety leads to negative immigrant attitude polarization, but only among those who reported not feeling happiness during the group discussions. Finally, both anger among those who reported group identity (in Figure 6), and anger among those who reported feeling anxious (Figure 7), led to greater policy difference polarization.

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TABLES AND FIGURES

Table 1: Emotional and Identity-Based Predictors of Behaviors in a Deliberative Poll

Independent Variable	Knowledge _{t2} : OLS [A]	Knowledge _{t2} : OLS [B]	Comments _{t2} : Logit [C]
Dependent variable _{t1}	.15 (.14)	.22 (.15)	10.45 (3.44)***
Happy	.54 (.19)***	1.27 (.46)**	-4.84 (1.89)**
Anxiety	.47 (.21)**	1.89 (.78)**	1.97 (.97)**
Anger	-.02 (.21)	-.09 (.23)	-1.10 (.77)
Group Identity	-.22 (.27)	-.19 (.25)	4.42 (2.01)**
Happy X Anxiety		-.51 (.29)	
Year in school	.39 (.14)**	.36 (.14)**	-.90 (.94)
Sex (male)	-.05 (.26)	-.07 (.25)	-6.89 (2.1)***
Age	.01 (.02)	-.00 (.02)	.58 (.23)**
White	.40 (.38)	.45 (.41)	-1.43 (1.76)
Party identification (Strong Democrat as high)	-.02 (.07)	-.03 (.07)	-1.02 (.42)**
Ideology	.01 (.06)	.04 (.06)	-.20 (.37)
Intercept	.46 (1.30)	-1.50 (1.68)	-.12 (7.27)
R ²	.36	.40	
Pseudo-R ²			.58
N	40	40	40

***p<.01, **p<.05, *p<.10

Table 2. Emotional and Identity-Based Predictors of Attitudes in a Deliberative Poll: OLS

Independent Variable	Negative Immigrant Attitudes Polarization _{t2} [D]	Negative Immigrant Attitudes Polarization _{t2} [E]	Policy Difference Polarization _{t2} [F]	Policy Difference Polarization _{t2} [G]
Dependent variable _{t1}	.42 (.12)***	.42 (.12)***	.08 (.17)	.09 (.17)
Happy	-.09 (.10)	.28 (.22)	-.21 (.24)	-.04 (.23)
Anxiety	1.15 (.39)***	.87 (.34)**	.29 (.22)	-.41 (.51)
Anger	-.11 (.06)*	-.12 (.07)	-.57 (.38)	-.42 (.44)
Group Identity	.74 (.25)***	.20 (.13)	-.08 (.37)	.35 (.30)
Anxiety X Group Identity	-.39 (.14)***			
Happy X Anxiety		-.28 (.11)**		
Anger X Group Identity			.40 (.16)**	
Anger X Anxiety				.46 (.23)*
Year in school	.11 (.11)	.10 (.11)	-.19 (.21)	-.17 (.21)
Sex (male)	-.30 (.20)	-.33 (.20)	.25 (.22)	.35 (.24)
Age	-.01 (.01)	-.01 (.01)	.05 (.02)***	.04 (.02)**
White	-.12 (.21)	-.10 (.20)	-.25 (.27)	-.04 (.30)
Party identification (Strong Democrat as high)	-.03 (.05)	-.02 (.04)	-.09 (.09)	-.04 (.09)
Ideology	-.05 (.04)	-.03 (.04)	-.08 (.08)	-.04 (.10)
Intercept	-.35 (.72)	.03 (.59)	.48 (1.72)	-.19 (1.61)
R ²	.58	.54	.45	.43
N	40	40	40	40

***p<.01, **p<.05, *p<.10

Table 3. Measures of Model Fit: Improvement by Including Identity and Affect

Dependent Variable	Model Fit ¹	Statistically Significant ²
<i>Knowledge</i>		
Basic model	.16	
Affect and Identity Included: Column A	.36	p<.10
<i>Typed comments</i>		
Basic model	16.58	
Affect and Identity Included: Column B	18.95	Not sig
<i>Negative Immigrant Attitudes Polarization</i>		
Basic model	.46	
Affect and Identity Included: Results not reported	.51	Not sig
Affect and Identity Interaction Included: Column D	.58	p<.10
<i>Policy Difference Polarization</i>		
Basic model	.27	
Affect and Identity Included: Results not reported	.37	Not sig
Affect and Identity Interaction Included: Column F	.45	P<.10

¹R², but for Typed Comments, Wald Chi-squared statistic

² Significant difference from basic model, two-tailed test

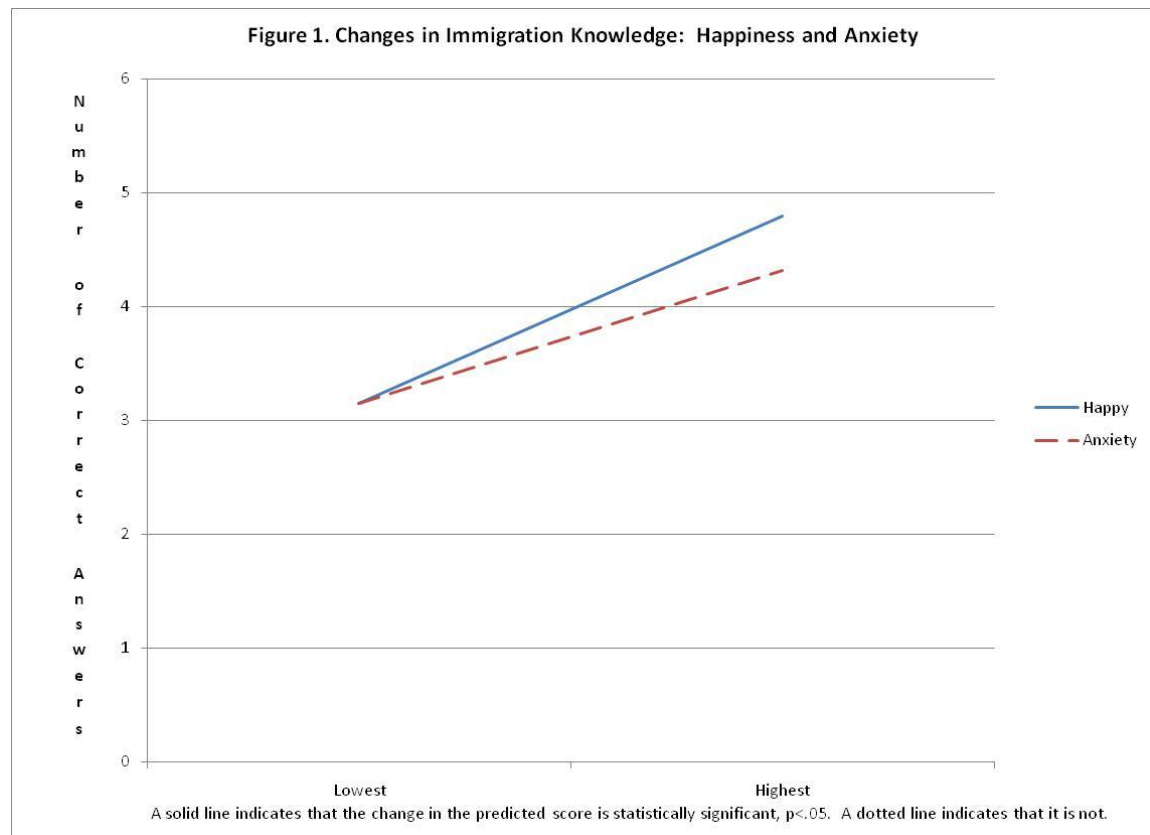


Figure 2. Changes in Immigration Knowledge: Happiness and Anxiety

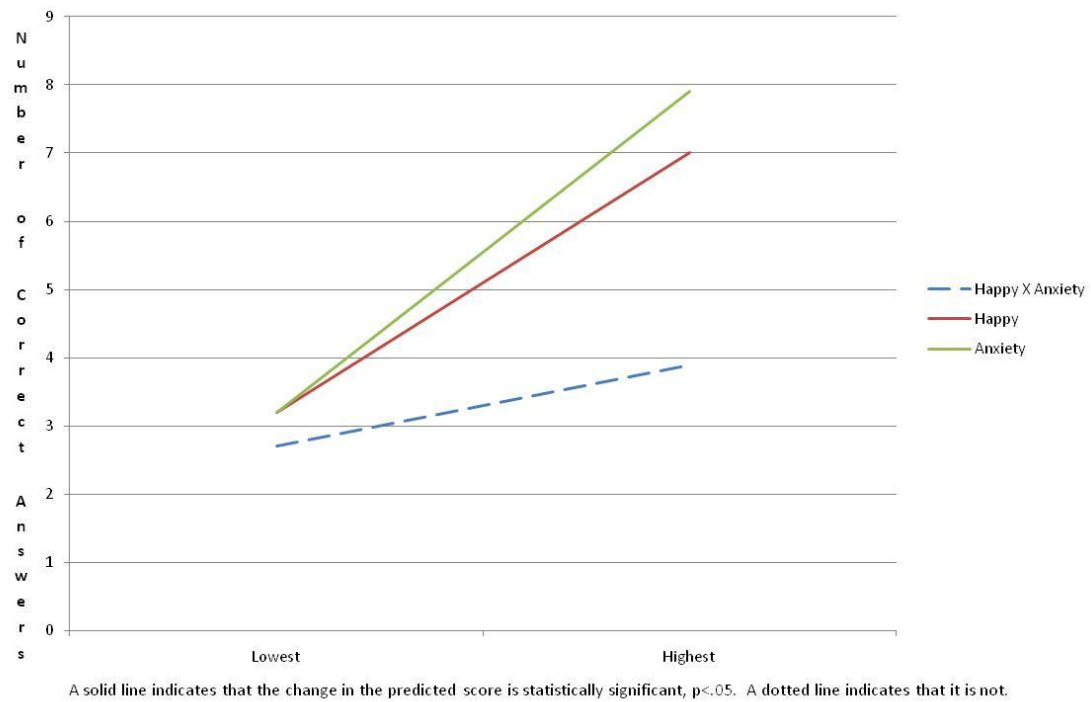


Figure 3. Predicted Probability of Typing a Comment

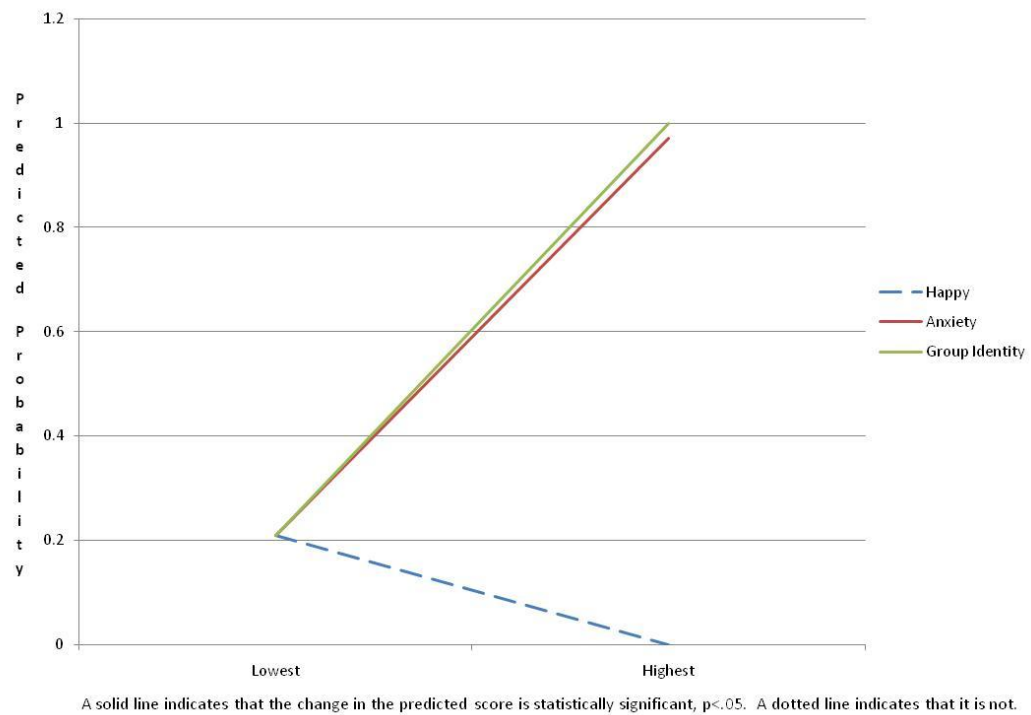


Figure 4. Negative Immigrant Attitudes Polarization: Anxiety and Group Identity

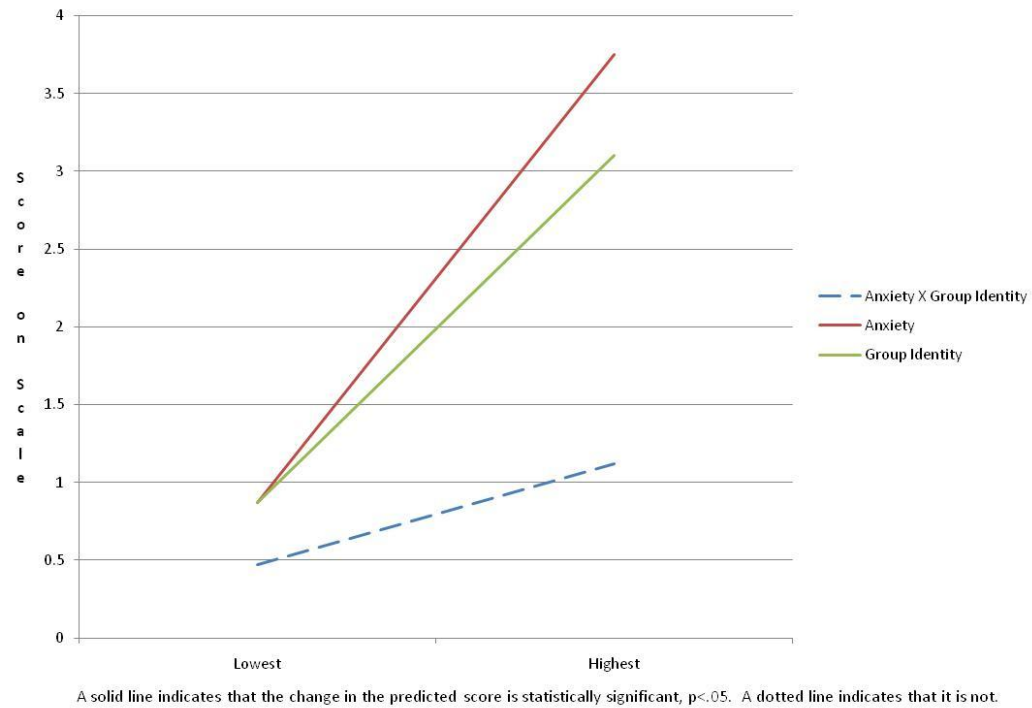


Figure 5. Negative Immigrant Attitudes Polarization: Happy and Anxiety

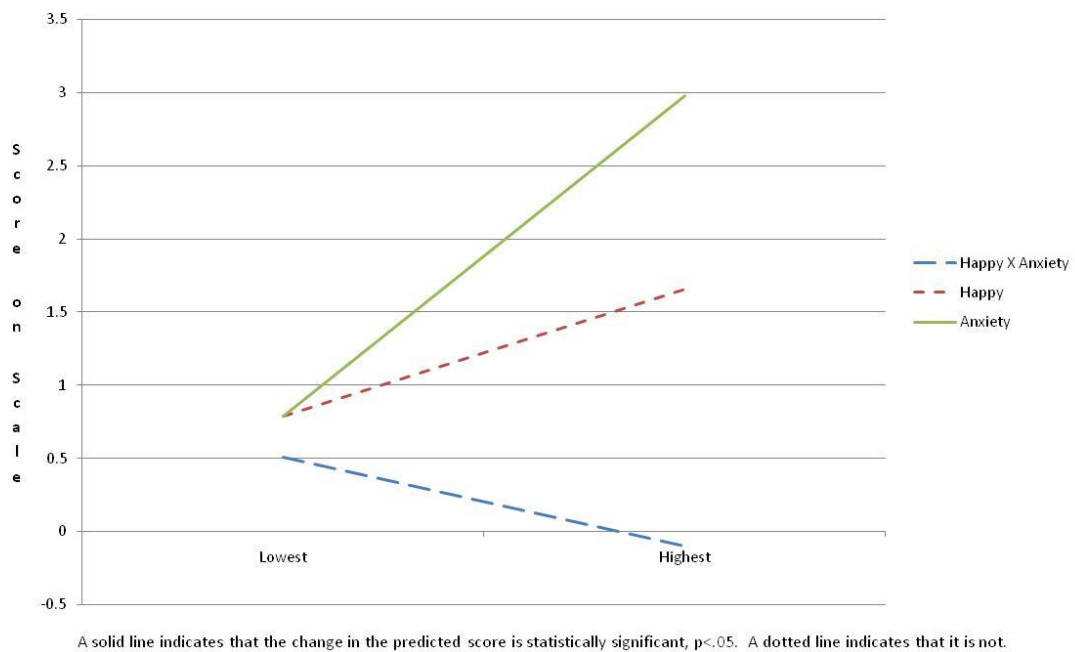


Figure 6. Policy Difference Polarization: Anger and Group Identity

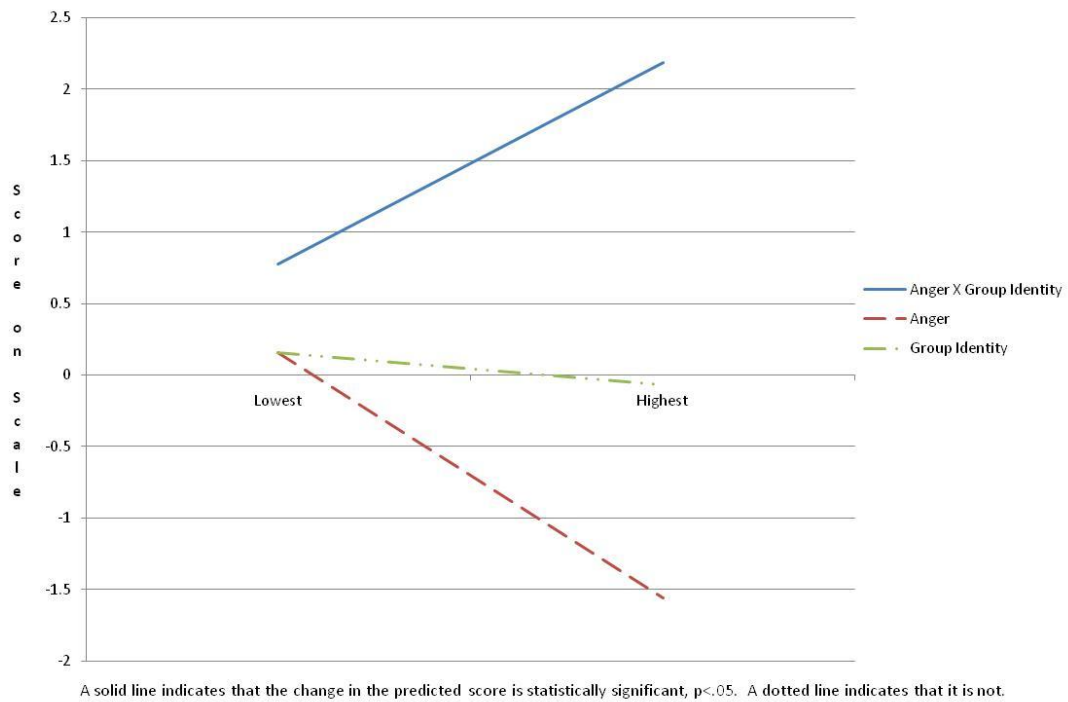


Figure 7. Policy Difference Polarization: Anger and Anxiety

